

FINAL REPORT

TEAM 7

PROJECT OVERVIEW , FLOW ,
PLAN AND RESULTS

Project Description

The Smart Nursery Guardian is an AI-assisted embedded system designed to continuously monitor and respond to an infant's needs in real time. The system integrates multiple advanced technologies, including machine learning, audio signal processing, environmental sensing, and embedded hardware control, to function as an intelligent caregiving assistant.

The system operates by continuously capturing and analyzing the infant's audio signals. Upon detecting a crying event, the system processes the input and classifies the type of cry using a trained machine learning model. Based on the classification outcome, appropriate actions are triggered, such as soothing mechanisms, alert generation, or caregiver notification. In parallel, the system continuously monitors environmental conditions—including motion, light intensity, temperature, and air quality—to ensure a safe and comfortable environment for the infant.

System Architecture

The system follows a distributed architecture and is divided into two main components:

1. Laptop-Based Application (Python)

The laptop is responsible for handling computationally intensive tasks, including:

- Capturing audio input via a microphone

- Detecting and processing infant cries

- Running the machine learning model for cry classification

Displaying system status and outputs through a Graphical User Interface (GUI)

Sending control commands to the microcontroller

2. Microcontroller Unit (Embedded System)

The microcontroller is responsible for real-time sensing and actuation, including:

Reading data from sensors (motion, light, temperature, gas)

Controlling actuators (servo motor, buzzer, LED, cooling fan)

Executing commands received from the laptop

System Components Interaction

Cry Detection and Classification

The system continuously monitors ambient audio. When a crying signal is detected:

Noise filtering techniques are applied

Voice Activity Detection (VAD) confirms the presence of a valid cry

Relevant features are extracted and passed to the machine learning model

The cry is classified into one of the following categories:

Hungry

Tired

Discomfort

Servo Motor (Rocking Mechanism)

Activated immediately upon cry detection

Simulates crib movement to soothe the infant

Stops when:

The crying ceases, or

A classification-based response is triggered

Buzzer (Alert System)

Activated in emergency situations

Turns ON when gas or smoke is detected

Remains OFF under normal operating conditions

LED (Lighting Control System)

The lighting system operates based on combined logical conditions:

Turns ON only when:

The environment is dark, AND

The infant is awake or crying

Turns OFF when:

The environment is sufficiently illuminated, OR

No motion or crying is detected

Motion Detection (PIR Sensor)

Detects movement near the crib

The infant is considered awake when:

Four or more motion events occur within an 8-second interval

This mechanism helps eliminate false triggers caused by minor disturbances

Temperature Monitoring and Cooling System

The system automatically regulates temperature using a fan:

Below 25°C → Fan OFF, Blue indicator

Between 25°C and 30°C → Fan operates at medium speed, Green indicator

Above threshold → Fan operates at maximum speed, Red indicator

This subsystem operates independently of the laptop to ensure real-time responsiveness.

Gas Detection (Safety System)

This subsystem has the highest priority in the system.

When gas or smoke is detected:

The buzzer is activated continuously

A critical alert is displayed on the GUI

A notification is sent to the caregiver via Telegram

System Operation Flow

Startup Phase

The microcontroller initializes all sensors and peripherals

The laptop application starts and loads the trained AI model

The GUI is launched, and audio monitoring begins

Continuous Operation

Normal State (No Cry Detected):

The system remains in monitoring mode, continuously analyzing environmental and audio inputs.

Cry Detection State:

Cry signal is detected and validated

Servo motor is activated immediately

The AI model classifies the cry

A corresponding response is executed

Response Based on Cry Classification

Hungry: Display notification and play calming media

Tired: Activate buzzer as an urgent alert

Discomfort: Display guidance message

Additional System Logic

Motion Handling:

The infant is considered awake once the motion threshold is reached

Lighting Control:

LED activation depends on darkness and infant activity

Emergency Handling:

Gas detection overrides all system operations and triggers emergency mode

Key Design Concept

The system is based on a distributed architecture model, where:

The laptop is responsible for intelligence, data processing, and decision-making

The microcontroller is responsible for real-time sensing and actuation

This design reflects a real-world Internet of Things (IoT) paradigm, ensuring both high computational efficiency and immediate responsiveness, which are critical for safety-critical applications such as infant monitoring systems.

Project Methodology and Development Plan

The development of the Smart Nursery Guardian system followed a structured, multi-phase methodology that integrates system simulation, hardware implementation, and intelligent software development. The workflow was carefully planned to ensure gradual validation and risk minimization at each stage of the project, in addition to maintaining continuous coordination through both offline and online meetings.

At the initial stage, the team focused on system modeling and simulation using Proteus Design Suite. The objective of this phase was to design and verify the complete hardware architecture in a controlled virtual environment before proceeding to physical implementation. The STM32F103C8 microcontroller was configured as the core processing unit, interfaced with multiple sensors including the MQ-2 gas sensor and PIR motion detector, as well as actuators such as motors, LEDs, and buzzers. Special consideration was given to circuit protection by implementing voltage divider networks to ensure compatibility between 5V sensor outputs and the 3.3V logic of the microcontroller. Additionally, a virtual debugging environment was established to monitor real-time system behavior and validate control logic execution .

Following the successful validation of the simulated system, the project transitioned into the hardware integration phase. During this stage, the team conducted **two online meetings** in a dedicated workspace. These sessions were scheduled during the hardware assembly period and were essential for building the physical prototype, interfacing all components, and performing hands-on testing and debugging. The team successfully verified multiple subsystems including motion detection, gas sensing, motor control, and alarm triggering. In parallel, the PCB design process using Altium Designer was initiated, aiming to transform the validated schematic into a manufacturable board layout .

In addition to the offline work, the team conducted **four offline (in-person) meetings** throughout the project timeline. These meetings were distributed across the different development phases and were primarily focused on coordinating software development, discussing system logic, assigning tasks, and tracking progress. They also played a key role in addressing technical challenges such as sensor calibration, communication protocols, and preparing for AI integration.

In the final stage of development, the system reached a high level of completion in both hardware and software domains. The hardware assembly was finalized, and all components were tested to ensure proper functionality. The software team also completed the development of both the AI model and the Graphical User Interface (GUI).

During the **latest meeting (held today)**, which represents the **sixth meeting overall (2 online + 4 offline)**, the team conducted practical testing of both the AI algorithms and the GUI system. The implemented codes were successfully executed, and both the AI model and the GUI operated correctly as intended. This meeting served as a critical validation milestone, confirming that the software components are functioning properly and are ready for full system integration. Despite this progress, minor issues such as serial communication are still under investigation to ensure seamless interaction between hardware and software components .

In summary, the project followed a clear and iterative development approach:

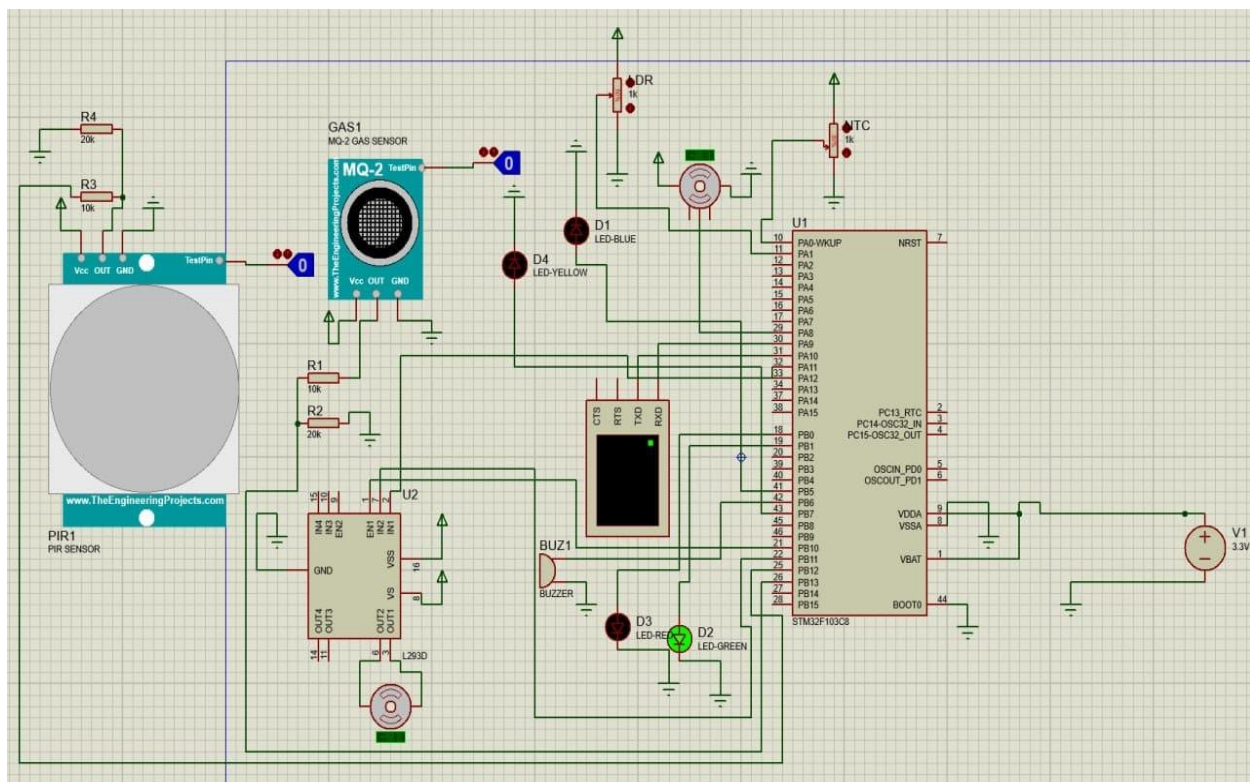
System Simulation → Hardware Implementation → Software & AI Development → System Integration and Testing, supported by continuous collaboration through **six total meetings (two offline and four online)** that ensured effective progress and coordination across all phases.

Simulation and Early Validation (Proteus)

As an additional development step, the team conducted a comprehensive system simulation using Proteus Design Suite prior to hardware implementation. This simulation phase played a crucial role in validating the system design, testing component interactions, and ensuring the correctness of the control logic in a risk-free virtual environment.

The Proteus simulation included the integration of the STM32 microcontroller with all essential sensors and actuators, allowing the team to emulate real operating conditions. It also enabled real-time debugging and monitoring of system behavior, which significantly reduced potential hardware issues during the implementation phase.

This step contributed to accelerating the development process, minimizing design errors, and providing a reliable foundation for the transition from simulation to physical system integration .



The project was successfully completed according to the planned timeline, with all major components developed, tested, and validated within their respective phases. The adopted methodology ensured steady progress and allowed the team to achieve each milestone without significant delays.

From a hardware perspective, all system components, including sensors and actuators, were fully assembled and tested. The system demonstrated reliable performance in detecting environmental changes such as motion and gas presence, and in executing the corresponding control actions. The hardware integration phase confirmed that the designed architecture operates efficiently under real-world conditions.

On the software side, both the Artificial Intelligence (AI) model and the Graphical User Interface (GUI) were successfully developed and finalized. The AI model achieved its intended functionality in processing and classifying input data, while the GUI provided an effective and user-friendly interface for system interaction .

During the final testing phase, conducted in the most recent meeting, the team verified the correct operation of both the AI and GUI systems. The implemented codes were executed successfully, and the outputs matched the expected behavior. This confirmed that the software components are functioning properly and are ready for integration with the hardware system.

Overall, the project outcomes indicate that all planned objectives have been achieved. The system components were delivered on time, functioned as expected, and collectively contributed to the successful realization of the Smart Nursery Guardian system. The project can therefore be considered a successful implementation of an integrated smart system combining hardware, software, and AI technologies.